

The Program in Action

When you run *digitalstream.py*, the output will look like this:

Digital Stream Screensaver, by Al Sweigart al@inventwithpython.com
Press Ctrl-C to quit.

```

                                0
                                0
                                0
1          0  0  1          1 0          1
0          0  0  1          0  0  0          0
0          1  0  0          0  1  0  0          1          0  1
0          1  0  0          1  011  1          1          0  1  0
0          1  0  0          0  000  11          0          0  1  1  0
1  1      0  1  0  1          1          110  10  1  0          1  0  1  0
          1  101  0          0          1          000  11  1  1          11  1  1  1
          0  100  1          0          11  00  0  1          01          0
          1  1      001  1          1          0  1  10  0          10          0
          0  0      010  0          1          1  11  11          0          0
--snip--
```

How It Works

Like Project 15, “Deep Cave,” this program uses the scrolling caused by `print()` calls to create an animation. Each column is represented by an integer in the `columns` list: `columns[0]` is an integer for the leftmost column, `columns[1]` is an integer for the column to the right of that one, and so on. The program initially sets these integers to 0, meaning it prints ' ' (an empty space string) instead of a stream in that column. Randomly, it changes each integer to a value between `MIN_STREAM_LENGTH` and `MAX_STREAM_LENGTH`. That integer decreases by 1 each time a line is printed. As long as a column’s integer is greater than 0, the program prints a random 1 or 0 in that column. This produces the “digital stream” effect you see on the screen.

```
1. """Digital Stream, by Al Sweigart al@inventwithpython.com
2. A screensaver in the style of The Matrix movie's visuals.
3. View this code at https://nostarch.com/big-book-small-python-projects
4. Tags: tiny, artistic, beginner, scrolling"""
5.
6. import random, shutil, sys, time
7.
8. # Set up the constants:
9. MIN_STREAM_LENGTH = 6 # (!) Try changing this to 1 or 50.
10. MAX_STREAM_LENGTH = 14 # (!) Try changing this to 100.
11. PAUSE = 0.1 # (!) Try changing this to 0.0 or 2.0.
12. STREAM_CHARS = ['0', '1'] # (!) Try changing this to other characters.
13.
14. # Density can range from 0.0 to 1.0:
15. DENSITY = 0.02 # (!) Try changing this to 0.10 or 0.30.
16.
17. # Get the size of the terminal window:
18. WIDTH = shutil.get_terminal_size()[0]
```